What Can Machine Learning Predict in NASDAQ Markets?

Price, Direction and Volatility Forecasting Across Models and Timeframes

Alessandro Pacifici
AI, Machine Learning and Big Data for Finance and Management · Waseda University
Final report · November 2026 · E-mini Nasdaq-100 futures (NQ), 2006–2025

Executive Summary

Volatility is predictable on every horizon tested; returns and direction show no useful improvement

Problem Markets are hard to forecast, but are all their characteristics equally hard? For the next NQ futures candle we ask what machine learning can predict: the return, the direction or the realized volatility.

Volatility

PREDICTABLE

−40%

squared log-RV forecast error vs persistence (DNN 128-64-32, 5 min)

vs the HAR-OLS benchmark: −6.8% at 5 min, −26% at 1 hour; rise or fall right 72% (majority 51%)

Direction

MARGINAL

+0.69 pp

accuracy over the majority class: 50.59% vs 49.90% (5 min)

≈ 7 extra right calls per 1,000 candles; seen in 2021–2023, not in 2024–2025; smaller or nil on longer candles

Return

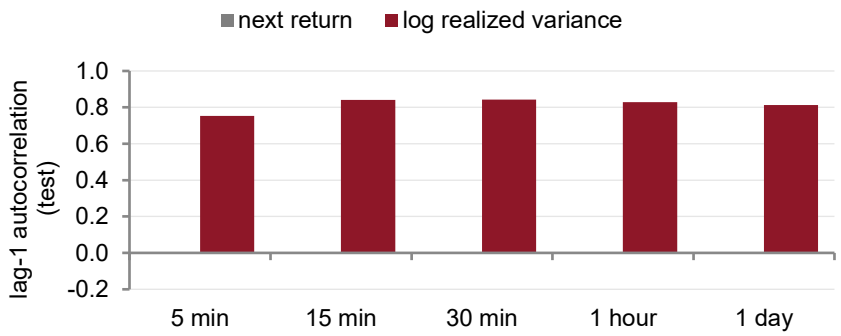
NOT DEMONSTRATED

R^2 −0.04%

vs forecasting zero, 95% interval [−0.10%, +0.02%] (5 min)

Every interval includes zero on all five candle lengths, from 5 minutes to 1 day

Why: volatility clusters, returns do not



Data & method

NQ futures, 1-minute data 2006–2025, 5 candle lengths (5 min–1 day); 15 models, benchmarks to deep networks and Mistral's Timestral. Train 2006–17, choose 2017–21, test 2021–25.

Originality

Experimental: one pipeline and test period for three targets and five horizons, compared like for like. Applied: ML and a new foundation model against the classical HAR benchmark.

Usefulness

Volatility forecasts are inputs to risk limits and position sizing; the study shows on which horizons ML adds to HAR, and where it adds nothing.

Implication

Use ML for realized volatility, not for next-candle returns or direction, with the models, data and horizons tried.

Exploratory: the test period (2021-07 → 2025-05; daily to 2025-01) was examined in earlier phases and the volatility selection score was revised after a first look at test results. Statistical accuracy only: no trading, costs or profitability were evaluated. pp = percentage points, bp = 0.01%, RV = realized variance.

© Alessandro Pacifici · Waseda University

Data & Methodology

One instrument, five candle lengths, three targets, one chronological split

NASDAQ historical data
NQ 1-min, 2006–2025

→ **Feature engineering**
lags, volatility, time of day

→ **Models**
benchmarks · ML · DNN · Timestral

→ **Forecasts**
for candle t+1

→ **Out-of-sample evaluation**
test from 2021-07 (exploratory)

Three targets for the next candle (t+1), decided at the close of t

Target	Definition	Main metric
1 Next return	r = log(C(t+1)/C(t))	R^2 vs the zero forecast
2 Next-candle direction	UP if C(t+1) > O(t+1); doji = miss	accuracy vs the majority class
3 Realized volatility	RV = Σ (1-min log returns)² in t+1	log-RV error vs persistence and HAR-OLS

- **Data:** E-mini Nasdaq-100 futures (NQ, CME), back-adjusted 1-minute series. No other market, no VIX.
- **Split:** train → 2017-08-24, validation → 2021-07-14, test 2021-07-15 → 2025-05-30 (daily test ends 2025-01-30, appendix A3).
- **No look-ahead:** features use candles ≤ t; scaling fitted on training data; labels never cross a session or a split.
- **Uncertainty:** 95% intervals from a moving-block bootstrap in 20-session blocks (Künsch, 1989).

Data profile, test sample

Candle	Test rows	UP / DOWN	ACF return	ACF log RV
5 min	273k	50% / 48%	−0.005	**+0.75**
15 min	90k	51% / 48%	+0.003	**+0.84**
30 min	44k	51% / 48%	−0.002	**+0.84**
1 hour	22k	52% / 48%	−0.000	**+0.83**
1 day	642	54% / 46%	−0.001	**+0.81**

Reading: candle direction is close to a coin flip; the lag-1 autocorrelation (ACF) of returns is about zero (−0.005 to +0.003), while log realized variance is strongly persistent (0.75 to 0.84).

23 features: past returns, momentum, rolling volatility, candle shape, RSI, moving-average gaps, volume, time of day, past log RV (1, 5, 22 candles). Timestral sees the last 512 closes.

© Alessandro Pacifici · Waseda University

What Is Predictable, and on Which Horizon?

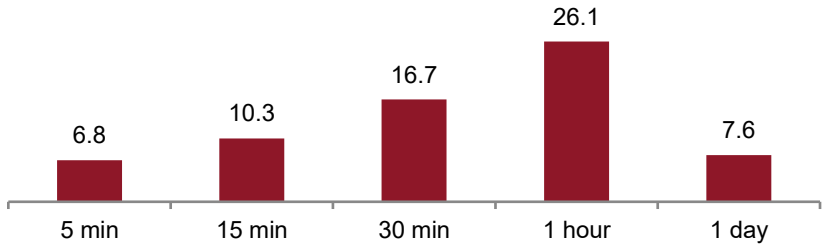
Only volatility beats its benchmark on every horizon; its gain over HAR-OLS grows with candle length

Task / metric (test)	5 min 273k candles	15 min 90k candles	30 min 44k candles	1 hour 22k candles	1 day 642 candles
Return R^2 vs zero (%)	**−0.04%** [−0.10, +0.02] DNN 256-128-64-32	**−0.01%** [−0.07, +0.04] XGBoost	**+0.00%** [−0.02, +0.02] XGBoost	**−0.01%** [−0.06, +0.04] DNN 128-64-32	**−1.95%** [−6.11, +1.73] XGBoost
Direction accuracy gain vs majority (pp)	**+0.69 pp** [+0.40, +0.98] DNN 256-128-64-32	**+0.25 pp** [−0.07, +0.57] DNN 64-32	**+0.16 pp** [−0.12, +0.38] DNN 256-128-64-32	**+0.26 pp** [−0.08, +0.55] XGBoost	**−0.78 pp** [−1.52, +0.00] DNN 128-64-32
Volatility squared log-RV error cut vs persistence (%)	**+40.0%** [+39.5, +40.4] DNN 128-64-32	**+36.5%** [+35.1, +37.7] DNN 256-128-64-32	**+42.6%** [+40.8, +44.3] DNN 256-128-64-32	**+52.0%** [+50.0, +54.1] XGBoost	**+18.7%** [+9.7, +24.5] DNN 128-64-32
Volatility squared log-RV error cut vs HAR-OLS (%)	**+6.8%** [+6.2, +7.3] DNN 128-64-32	**+10.3%** [+8.8, +11.6] DNN 256-128-64-32	**+16.7%** [+14.5, +18.7] DNN 256-128-64-32	**+26.1%** [+23.8, +28.5] XGBoost	**+7.6%** [−0.9, +14.6] DNN 128-64-32

95% interval above zero interval includes zero interval below zero

- **Volatility:** every horizon beats persistence (squared log-RV error 19% to 52% lower). The extra gain over HAR-OLS rises from 6.8% (5 min) to 26.1% (1 hour); on daily candles the interval includes zero.
- **Return and direction:** no interval is above zero except 5-minute direction (+0.69 pp).

Volatility: squared log-RV error cut vs HAR-OLS (%)



Each cell: the trained model with the highest validation z for that task and candle length, scored on the test. Daily sample: 642 test sessions (appendix A3).

© Alessandro Pacifici · Waseda University

All Models Tested

15 models on 5 timeframes: 150 model–timeframe–task configurations, all run on the same data

Family	Model	Method	Tasks
Baselines	**Zero return / training mean**	naive forecast	R
	Majority class	most frequent training class	D
	Persistence · rolling mean (22)	RV(t+1) = RV(t) · average past variance	V
Statistical	**HAR-OLS**	regression on past log RV (1, 5, 22 candles) — Corsi (2009)	V
	Ridge / logistic regression	L2-penalised, strength chosen on validation	R · D · V
Machine learning	**Random forest**	200 trees, depth 8	R · D · V
	XGBoost	gradient boosting, early stopping	R · D · V
Deep learning	**DNN: 64-32-16 · 64-32 · 128-64-32 · 256-128-64-32**	feed-forward, ReLU, AdamW, early stopping	R · D · V
Mistral AI	**Timestral-5M, frozen weights**	pretrained forecaster of the next close (zero-shot)	R · D
	Timestral-5M + calibration	forecast spread → log RV, linear map fitted on validation	V

Notes

- **R** return · **D** direction · **V** volatility
- **Mistral's role:** Timestral is a pretrained time-series model (local weights, no API); it is not a language model and reads no text.
- **Not run:** LSTM; larger Timestral sizes (28M–209M).
- **Every fit and hyper-parameter:** reports/multitarget_final/models.csv.

© Alessandro Pacifici · Waseda University

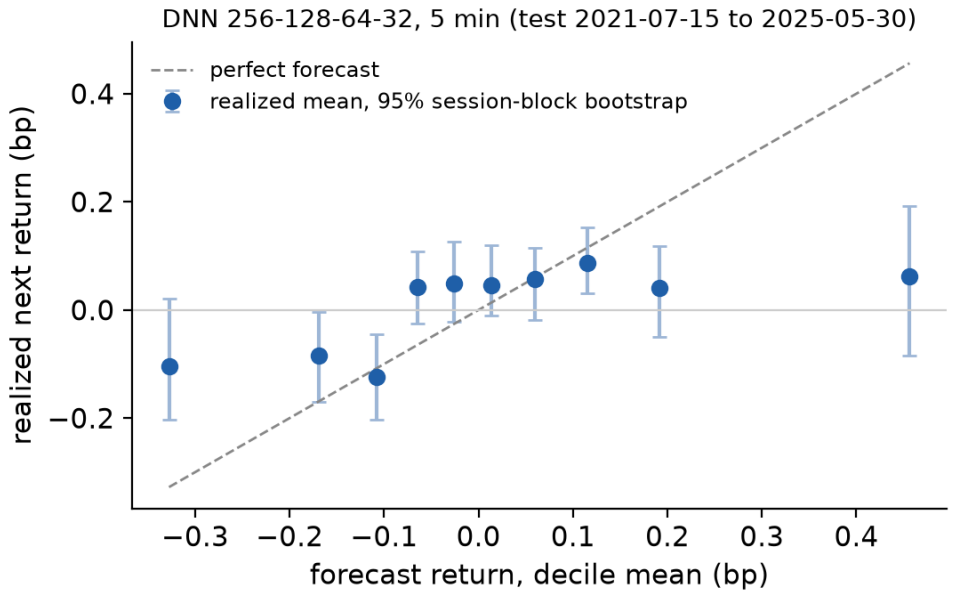
Price / Return Forecasting

Five models on 5 min candles: none improves on the zero-return forecast

#	Model	Val. z	RMSE bp	R^2 vs zero	95% interval (%)	Sign right
1	**DNN 256-128-64-32**	3.1	7.71	**−0.04%**	[−0.10, +0.02]	51.3%
2	**XGBoost**	2.5	7.71	**+0.00%**	[−0.02, +0.02]	51.2%
3	**Ridge**	0.8	7.71	**−0.03%**	[−0.09, +0.03]	51.2%
4	**Random forest**	0.1	7.71	**−0.05%**	[−0.09, −0.02]	51.2%
5	**Timestral-5M (frozen weights) ***	-4.3	7.91	**−2.55%**	[−3.43, −1.62]	50.2%

Benchmark (zero-return forecast): RMSE 7.71 bp on all 5 min test candles. * Timestral: scored on its fixed random sample of 20,000 test candles (its benchmark on the same candles; appendix A4).

- **Result:** the best R^2 is +0.00%; every interval includes or lies below zero, so no model beats "the next return is zero". The same holds on all five candle lengths (slide 4).
- **Predicted vs realized (right, #1):** realized return per forecast decile: forecasts span less than ±0.5 bp while a typical candle moves about 7.7 bp.
- **Reading:** linear, tree, deep and foundation models fail in the same way: no usable return signal with the models, data and horizons tried (consistent with, but not proof of, a random walk).



Rows ranked by validation evidence z (validation improvement over the task's naive benchmark ÷ its block-bootstrap standard error); all other columns are test results. R^2 interval: model and zero-forecast errors resampled together. Test 2021-07-15 → 2025-05-30.

© Alessandro Pacifici · Waseda University

Next-Candle Direction Classification

Five models on 5 min candles: 4 of 5 beat the majority class, by less than one percentage point

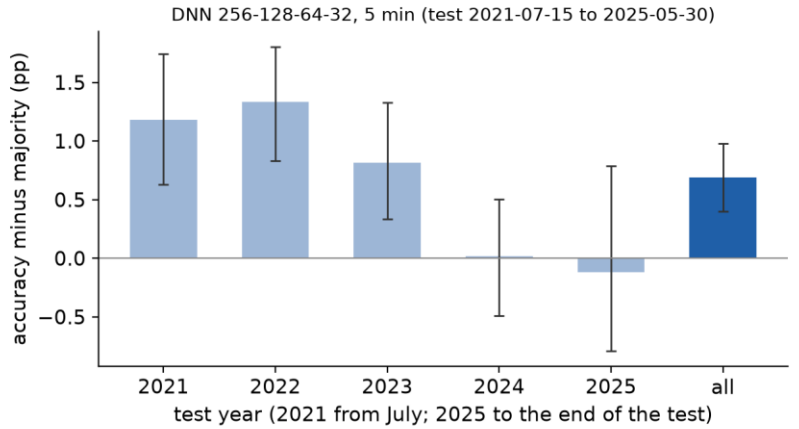
#	Model	Val. z	Accuracy	Majority	Gain	95% interval	Balanced acc.	F1 UP
1	**DNN 256-128-64-32**	11.9	**50.59%**	49.90%	**+0.69 pp**	[+0.40, +0.98]	51.37%	0.559
2	**Random forest**	11.0	**50.52%**	49.90%	**+0.62 pp**	[+0.32, +0.91]	51.36%	0.541
3	**Logistic**	11.0	**50.40%**	49.90%	**+0.50 pp**	[+0.24, +0.77]	51.23%	0.542
4	**XGBoost**	10.7	**50.47%**	49.90%	**+0.58 pp**	[+0.27, +0.89]	51.34%	0.535
5	**Timestral-5M (frozen weights) ***	-1.9	**49.25%**	49.84%	**−0.59 pp**	[−1.65, +0.45]	50.10%	0.517

* Timestral: scored on its fixed random sample of 20,000 test candles (its benchmark on the same candles; appendix A4). Accuracy counts dojis (open = close) as misses.

#1 DNN 256-128-64-32: confusion matrix (test)

	Actual UP	Actual DOWN
Predicted UP	**82,434 (30.7%)**	76,180 (28.4%)
Predicted DOWN	53,956 (20.1%)	**55,837 (20.8%)**

- **Size:** +0.69 pp ≈ 7 more right calls per 1,000 candles. Tree, linear and deep models are within a few tenths of each other; Timestral does not beat the majority class.



Stability of #1 by test year: gain above zero in 2021, 2022, 2023; interval includes zero in 2024, 2025.

© Alessandro Pacifici · Waseda University

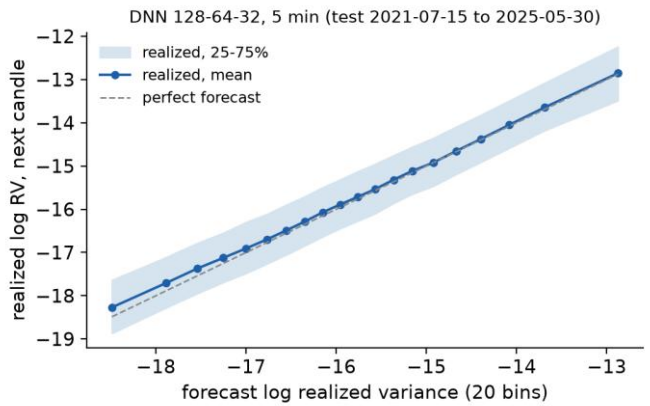
Volatility Forecasting

Five models on 5 min candles: all beat persistence; tree and deep models also beat the HAR-OLS benchmark

#	Model	Val. z	Error cut vs persistence [95%]	Error cut vs HAR-OLS [95%]	RMSE log RV	RV up/down right	Up/down vs HAR-OLS [95%]
1	**DNN 128-64-32**	44.0	**+40.0% [+39.5, +40.4]**	**+6.8% [+6.2, +7.3]**	0.901	**72.2%**	**+1.13 pp [+0.95, +1.31]**
2	**XGBoost**	42.4	**+42.0% [+41.5, +42.4]**	**+9.9% [+9.4, +10.3]**	0.885	**72.8%**	**+1.71 pp [+1.52, +1.88]**
3	**Random forest**	27.4	**+37.1% [+36.5, +37.6]**	**+2.3% [+1.7, +2.8]**	0.922	**71.4%**	**+0.37 pp [+0.25, +0.50]**
4	**Ridge**	23.1	**+35.7% [+34.9, +36.3]**	**+0.1% [−0.9, +0.9]**	0.932	**70.8%**	**−0.27 pp [−0.51, −0.05]**
5	**Timestral-5M + calibration ***	6.4	**+12.2% [+9.7, +14.8]**	**−35.9% [−38.8, −33.2]**	1.087	**67.7%**	**−3.36 pp [−4.02, −2.71]**

Benchmarks on all 5 min test candles: persistence (next RV = current RV) RMSE 1.163, up/down majority 51.1%; HAR-OLS RMSE 0.933, up/down 71.0%. * Timestral: scored on its fixed random sample of 20,000 test candles (its benchmark on the same candles; appendix A4).

- **Metric:** "error cut" = reduction of the squared log-RV forecast error vs a benchmark (an out-of-sample R^2). For #1 it is 40.0% vs persistence; the RMSE itself falls by 22.5% (0.901 vs 1.163).
- **HAR-OLS is the bar to clear:** it already captures clustering and the time-of-day pattern, so the cut vs HAR-OLS (6.8% for #1) is what ML adds. On the test XGBoost (#2 by validation z) is slightly ahead.
- **Scope:** realized volatility from 1-minute returns (Andersen & Bollerslev, 1998), not implied volatility; no option prices or option strategies were studied.



Rows ranked by validation evidence z (validation improvement over the task's naive benchmark ÷ its block-bootstrap standard error); all other columns are test results. Volatility score revised from QLIKE (Patton, 2011) to log-RV error after a first look at test numbers: not blind.

© Alessandro Pacifici · Waseda University

Which NASDAQ Market Characteristic Is Most Predictable?

Strongest evidence of improvement: 5 min realized volatility, DNN 128-64-32

How the configurations were ranked

1. Each model × timeframe is scored on VALIDATION data (2017–2021) by its improvement over its task's naive benchmark.
2. z = that improvement ÷ its bootstrap standard error: the strength of evidence against that benchmark, not a common scale of usefulness across tasks.
3. The highest z wins within each task (slides 6–8); across tasks z only says where the evidence of beating the naive benchmark is strongest.
4. Winners are then scored on the TEST block. The volatility score was revised after a first look at test numbers: the ranking is exploratory.

Task	Task winner (validation)	Val. z	Test vs naive benchmark
Volatility	DNN 128-64-32, 5 min	**44.0**	+40.0% error cut vs persistence [+39.5, +40.4]
Direction	DNN 256-128-64-32, 5 min	**11.9**	+0.69 pp vs majority [+0.40, +0.98]
Return	DNN 256-128-64-32, 5 min	**3.1**	R^2 −0.04% vs zero [−0.10, +0.02]

STRONGEST EVIDENCE OF IMPROVEMENT (HIGHEST VALIDATION z)

DNN 128-64-32 · 5 min realized volatility

−40.0%

squared log-RV forecast error vs persistence on the test, 95% interval [−40.4%, −39.5%]; validation z 44.0

−6.8%
squared error vs HAR-OLS [−7.3, −6.2]

−22.5%
RMSE of log RV vs persistence (0.901 vs 1.163)

72.2%
RV up/down right (majority 51.1%, HAR-OLS 71.0%)

On the test XGBoost (#2 by validation z) cuts the error slightly more (−42.0%); the ranking is kept as computed on validation, not re-chosen on the test. Longer candles show larger gains over HAR-OLS (up to −26.1% at 1 hour, slide 4).

Implications: realized volatility is the most predictable characteristic here, useful as an input to risk limits and sizing; for returns no improvement was demonstrated, and the direction gain is small and fades after 2023. **Limits:** exploratory test; one market; statistical accuracy only — commissions, spread and latency were not modelled, so economic value (e.g. of a +0.69 pp direction gain) is unknown.

© Alessandro Pacifici · Waseda University

Reproducibility & References

Every table, chart and slide is regenerated from code in the repository; the licensed NQ data are not published

Code, results and this deck https://github.com/Paciooo/Project_ai_ML_bigdata · branch main

How to reproduce (details: appendix A1)

Environment	Python 3.12 + requirements-lock.txt, Node.js 18+
Data	NQ 1-minute bars 2006–2025 (licensed, not in Git): set FUTURES_DATA_DIR
Run	multitarget.py → multitarget_report.py → verify_multitarget.py
Deck	presentation: npm ci; node build_deck.js

- **Checked:** verify_multitarget.py recomputes every reported point estimate, the rankings and the deck data from the saved forecasts (0 problems); unit tests cover targets, timing, bootstrap and the Timestral calibration.
- **Not re-checked:** the raw data, the training (not re-run) and the bootstrap intervals (order only).

References

Andersen, T. G., & Bollerslev, T. (1998). Answering the skeptics: Yes, standard volatility models do provide accurate forecasts. International Economic Review, 39(4), 885–905.

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. Journal of Financial Econometrics, 7(2), 174–196.

Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. Annals of Statistics, 17(3), 1217–1241.

Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. Journal of Econometrics, 160(1), 246–256.

Mistral AI (2026). Timestral: time series foundation models. Announcement mistral.ai/news/timestral; code github.com/mistralai/timestral (commit a534bdb); weights huggingface.co/mistralai/Timestral-5M.

© Alessandro Pacifici · Waseda University